



TEMPERATURE

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► Recall that $\beta \mapsto \log Z(\beta) \equiv: A(\beta)$ is a convex function.

▶ The entropy and log-partition functions are dual!

$$Z(\beta) = \int_{\mathbf{x}} \exp(-\beta U) \eta(x) \mathrm{d}x$$

$$A^*(u) = \sup_{\beta} [-\beta u - \log Z(\beta)]$$

$$\frac{\mathrm{d}}{\mathrm{d}\beta}(\beta u - A(\beta)) = 0 \quad \Longleftrightarrow \quad \frac{\mathrm{d}}{\mathrm{d}\beta} \log Z(\beta) = -u$$

$$\Leftrightarrow \langle U \rangle_{\beta} = u$$

$$A^*(v) \equiv -\beta \langle U \rangle_\beta - \log Z(\beta) \equiv S(v)$$

$$S(u) = \max_{\pi_\beta[U]=u} S(\pi_\beta)$$



For $v \in U(\infty)$ we define:

$$\log Z(\beta) \equiv \inf_u [-\beta u + S(u)]$$

$$S(u) = \sup_{\beta} [\beta u + \log Z(\beta)]$$

There partition function, and entropy encode the same information!

TEMPERATURE

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- ▶ Recall that $\beta \mapsto \log Z(\beta) =: A(\beta)$ is a convex function.

$$Z(\beta) = \int_{\mathbb{X}} \exp(-\beta U) \eta(x) dx$$

- ▶ For $u \in U(\mathbb{X})$ we define:

$$S(u) = \max_{\pi_\beta[U]=u} S(\pi_\beta)$$

- ▶ The entropy and log-partition functions are dual!

$$\log Z(\beta) = \inf_u [-\beta u + S(u)]$$

$$S(u) = \sup_\beta [\beta u + \log Z(\beta)]$$

- ▶ Therefore partition function, and entropy encode the same information!

FREE ENERGY